

Title: Solving routing problems using quantum annealing

Expected time availability of the interns: at least 20 hours per week.

Expected number of interns: 5

Requirements: experience with quantum annealing, Leap framework, and designing QUBO formulations for combinatorial optimization problems. Experience with combinatorial optimization, probabilistic methods, QUBO formulations, and foundational quantum computing concepts is expected. Nice-to-have: experience with variational quantum algorithms and machine learning.

The project's goal is to investigate quantum annealing approaches to solve the Capacitated Vehicle Routing Problem with Time Windows (CVRPTW). We've identified several suitable QUBO formulations for this problem, but would like to reduce the number of necessary qubits, implement the solution, and conduct the experiments using D-Wave's Leap framework. One of the options to achieve it is through graph coarsening techniques. This project will also investigate exploration dynamics across classical and hybrid quantum-classical optimization approaches. It will also explore machine learning techniques to find good values of the relaxation parameters.

Plan for the internship:

1. In the first week, the interns will familiarize themselves with QUBO formulations for CVRPTW proposed in this project and with the relevant research papers.
2. In the next 2-3 weeks, they will work on designing and implementing the best techniques.
3. In the last 2-3 weeks, the interns will finalize the implementation and conduct experiments on benchmark datasets, like Solomon dataset.
4. The implementations will be available in a publicly available repository under an open-source license. Depending on the results, we will also consider the preparation of a research publication after the QIntern program.

In terms of exploring dynamics in hybrid quantum-classical optimization for constrained combinatorial problems, the interns will follow this plan:

https://drive.google.com/file/d/1CuYalkLHt_PzcOCjGluJF_Sox_YyWL3I/view?usp=sharing

Some other relevant sources:

1. Holliday J.B., Blount D., Osaba E., and Luu K., "Advanced Quantum Annealing Approach to Vehicle Routing Problems with Time Windows", arXiv:2503.24285, 2025.
2. "Solving the CVRPTW: Quantum Annealing vs. Exact Optimization", Master's Thesis by Oscar Axel Vargas Salomón, 2019.

3. Transportation Optimization Portal of SINTEF Applied Mathematics, VRPTW dataset:
<https://www.sintef.no/projectweb/top/vrptw>

4. Mustafa Mert Özyılmaz, "Graph-Coarsening Approach for the Capacitated Vehicle Routing Problem with Time Windows", <https://arxiv.org/abs/2510.22329>

5. K. Nałęcz-Charkiewicz, A. Das, T. Chatterjee, J. Keene, P. Gora and C. C. N. Kuhn, "Graph Coarsening Approach to the Vehicle Routing Problem: An Approximation Strategy," in IEEE Access, vol. 13, pp. 22459-22472, 2025, doi: 10.1109/ACCESS.2025.3534677:
<https://ieeexplore.ieee.org/document/10854442>.